

53. Maximum Subarray

Medium Topics Companies

Given an integer array `nums`, find the **subarray** with the largest sum, and return *its sum*.

Example 1:

Input: `nums = [-2,1,-3,4,-1,2,1,-5,4]`

Output: 6

Explanation: The subarray `[4,-1,2,1]` has the largest sum 6.

Example 2:

Input: `nums = [1]`

Output: 1

Explanation: The subarray `[1]` has the largest sum 1.

Example 3:

Input: `nums = [5,4,-1,7,8]`

Output: 23

Explanation: The subarray `[5,4,-1,7,8]` has the largest sum 23.

Constraints:

$1 \leq \text{nums.length} \leq 10^5$

$-10^4 \leq \text{nums}[i] \leq 10^4$

Follow up: If you have figured out the $O(n)$ solution, try coding another solution using the **divide and conquer** approach, which is more subtle.

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Code

Java Auto

```
1 class Solution {
2     public int maxSubArray(int[] nums) {
3         int currentSum = nums[0];
4         int maxSum = nums[0];
5
6         for (int i = 1; i < nums.length; i++) {
7             currentSum = Math.max(nums[i], currentSum + nums[i]);
8             maxSum = Math.max(maxSum, currentSum);
9         }
10
11         return maxSum;
12     }
13 }
```

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Ln 13, Col 2

Testcase Test Result

Accepted Runtime: 0 ms

Case 1 Case 2 Case 3

Input

nums =
[-2,1,-3,4,-1,2,1,-5,4]

Output

6

Expected

6